



United International University (UIU)

Dept. of Computer Science & Engineering (CSE)

Term Exam, Fall 2025
CSE 3811 : Artificial Intelligence
Total Marks: 40 Duration: 2 hours

Answer all the questions. Figures in the right-hand margin indicates full marks.

Any examinee found adopting unfair means will be expelled from the trimester / program as per UIU disciplinary rules.

1. An anime fan survey was conducted among 450 participants to study their preference for different anime themes. Fans were classified as university students and working professionals.

Out of the total participants:

- 270 were students and 180 were working professionals.
- Among the professionals, 150 were males.
- Among the students, 120 were females.

Each participant selected one favourite anime theme from **Action**, **Slice-of-Life**, or **Supernatural**. The preferences were recorded as follows:

- Among male professionals, 50 preferred **Action**, 40 preferred **Slice-of-Life**, and 60 preferred **Supernatural**.
- Among female professionals, 10 preferred **Action**, 10 preferred **Slice-of-Life**, and 10 preferred **Supernatural**.
- Among male students, 60 preferred **Action**, 45 preferred **Slice-of-Life**, and 45 preferred **Supernatural**.
- Among female students, 55 preferred **Action**, 40 preferred **Slice-of-Life**, and 25 preferred **Supernatural**.

- a) A randomly selected participant prefers **Supernatural**. What is the probability that the participant is a working professional ? [3]
- b) Find the probability that a randomly selected participant is a female student or an **Action** anime fan. [2]

2. A university library has four study rooms:

Room 1 (R1), Room 2 (R2), Room 3 (R3), and Room 4 (R4).

Six students — Imran (I), Bushra (B), Kamal (K), Nusrat (N), Rafi (R), and Tahsin (T) — want to be assigned study rooms during the same time slot. Each student will be assigned exactly one room.

You collect the following information:

- i. Imran does not want Room 4, as it is too noisy.
- ii. Bushra and Kamal want to study together, so they must be assigned to the same room.
- iii. Nusrat prefers quiet places, so she will only choose Room 1 or Room 2.
- iv. Rafi wants a different room from Tahsin, because they get distracted easily.
- v. Tahsin insists on being in the same room as Imran.
- vi. Room 3 can accommodate at most two students.
- vii. Imran wants to be in a different room from Bushra, because he needs extra space.
- viii. Tahsin refuses to be in Room 2, as it is too cold.
- ix. Rafi must be in Room 4, because he needs a computer setup only available there.

Now, formulate this situation as a Constraint Satisfaction Problem (CSP) and determine a valid room assignment for each student. You must solve the CSP using MRV and LCV heuristics. [5]

3. Answer the following questions.

- a) Sun corporation is a well-renowned company that sells batteries. The humidity of the factory affects the battery significantly. A battery is good 85% of the time. On any day, the humidity is low 92% of the time when the battery is good. Also, when the battery is bad, the humidity is low 13% of the time.

What is the probability that a battery is good when the humidity is high?

[3]

- b) A Bayesian network has been built for your local crop producing lands. Using the information gathered during the past few months, the conditional probability values have been acquired. Now, what is the probability that we will have high crop yield (+c) if there is no rainfall (-r) and the irrigation system is faulty (-i).

[5]

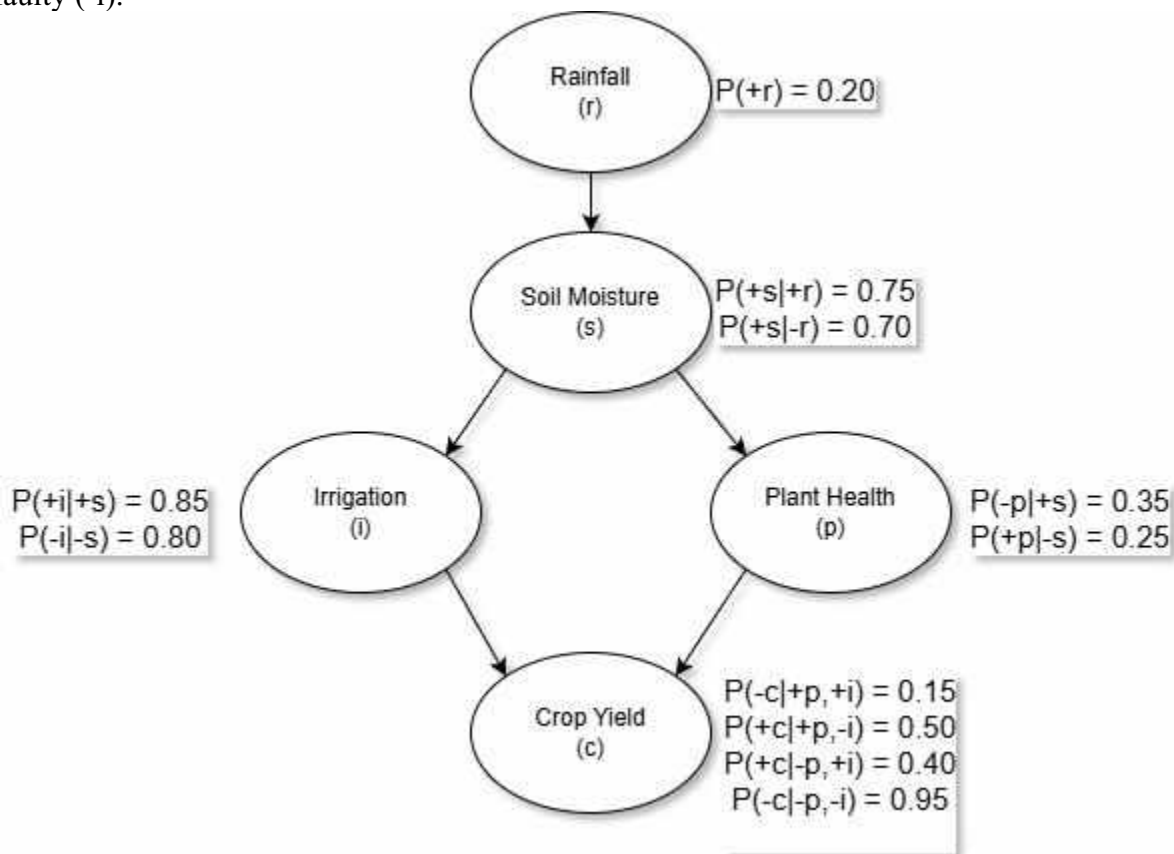


Figure 3(b)

- c) For the Bayesian network given in question-B, if the full joint distribution values of all the random variables were given instead of the conditional probability values, what would be the advantage and disadvantage?

[2]

4. A Coaching have been trying to improve the result of their student. In this regards they started to collect the some data from their last years batch. Suppose you have been hired as a data scientist to develop a Decision Tree model for this task. The dataset provided is as follows:

StudyTime	Attendance	Assignment	Sleep	Result
High	Regular	Complete	Good	Pass
High	Regular	Complete	Poor	Pass
Medium	Regular	Complete	Good	Pass
Medium	Irregular	Incomplete	Poor	Fail
Low	Regular	Incomplete	Poor	Fail
Low	Irregular	Incomplete	Poor	Fail
Medium	Regular	Incomplete	Good	Pass
High	Irregular	Complete	Good	Pass
Medium	Irregular	Complete	Poor	Fail
Low	Regular	Complete	Good	Pass

- a) Build the best decision tree using the above information. Show the calculation. [8]
b) How does (basic) decision tree model perform if the dataset have slight error? Show your reasoning. [2]

5. Suppose you are designing a classifier agent for an online shop that will be able to predict whether a customer will buy or not buy a product based on the customer's Age, Income, Marital status and Browser. The following data has been observed from history of previous customers:

ID	Age Group	Income	Marital Status	Browser	Class
1	Young	High	Single	Mobile	Buy
2	Young	Medium	Single	Desktop	Buy
3	Middle	High	Married	Desktop	Buy
4	Senior	Medium	Married	Mobile	Buy
5	Middle	Medium	Single	Mobile	Buy
6	Young	Low	Single	Mobile	Not Buy
7	Senior	Low	Married	Desktop	Not Buy
8	Middle	Low	Married	Mobile	Not Buy
9	Senior	Medium	Single	Desktop	Not Buy
10	Young	Medium	Married	Mobile	Buy
11	Middle	High	Single	Desktop	Buy
12	Senior	Low	Single	Mobile	Not Buy

- a) For a new data instance with **Age Group = Middle, Income = Medium, Marital Status = Single, Browser = Mobile**, determine if the customer is likely to buy or not buy using Naive Bayes Classifier. Use Laplacian smoothing with $k=1$. Show the calculations in detail. [8]
b) Explain briefly how the “zero-frequency problem” is handled in Naïve Bayes. [2]